

5.5 Assistive Technologies

معاون ٹیکنالوجیز (Assistive Technologies) وہ آلات، سافٹ ویئر اور نظام ہیں جو معذور افراد کی مدد کرتے ہیں تاکہ...

CS XI

Chapter 5: Application and Impacts of Computing

Lecture 10 of 12

★ Golden

5.5 Assistive Technologies ★ Golden

LEARN IT

Assistive technologies are **devices, software and systems** that help people with **disabilities or functional limitations** (difficulty in seeing, hearing, moving, speaking, remembering, etc.). They allow more people to **participate equally** in school, work and society.

There are **four categories**:

Category	Purpose	Examples
1. Mobility aids	Help people move from one place to another.	Smart wheelchairs, stair lifts
2. Sensory aids	Support vision and hearing to improve interaction.	Hearing aids, screen readers (VoiceOver), Braille displays
3. Communication aids	Assist individuals with speech difficulties to express themselves.	Speech-generating devices, text-to-speech apps
4. Cognitive / learning aids	Help people with memory, reading and understanding.	Audiobooks, spell-checkers, digital reminder apps

Mobility

· Sensory

· Communication

· Cognitive

- A **screen reader** is software that reads aloud the text on a screen, so a blind person can use a phone or computer. **VoiceOver** is Apple's screen reader; Android has TalkBack.
- A **Braille display** is a device with small pins that rise and fall to form Braille dots, so a blind user can read screen text with their fingers.
- A **speech-generating device** speaks out words that a person types or selects on a screen.
- A **smart wheelchair** can be controlled by a joystick, voice or head movement and may avoid obstacles with sensors.

WORKED EXAMPLE

Scenario — choose the right assistive technology:

(a) Ahmed, a student in Hyderabad, is blind and wants to read his e-textbook. → **Sensory aid**: a screen reader or Braille display.

(b) Mrs Rashida cannot climb the stairs of her house after a knee operation. → **Mobility aid**: stair lift (or smart wheelchair).

(c) Zain cannot speak clearly due to a medical condition but can type. → **Communication aid**: text-to-speech app or speech-generating device.

(d) Nani often forgets to take her medicine. → **Cognitive aid**: digital reminder app.

(e) Usman has dyslexia (difficulty reading). → **Cognitive/learning aid**: audiobooks and spell-checkers.

Method: identify the *limitation* (moving, seeing/hearing, speaking, memory/reading) → choose the matching category.

REMEMBER IT

Four categories = **"My Small Cat Cries": Mobility, Sensory, Communication, Cognitive.** Link each to a body function: *legs* (mobility), *eyes/ears* (sensory), *mouth* (communication), *mind* (cognitive).

COMMON MISTAKE

Mistake: Putting text-to-speech and screen readers in the same category every time. **Fix:** In the source, the **screen reader** is a *sensory* aid (supports vision), while a **text-to-speech app** is listed as a *communication* aid (helps a person with speech difficulty express themselves). Decide by *whose problem* it solves.

اردو میں سمجھیں

معاون ٹیکنالوجیز (Assistive Technologies) وہ آلات، سافٹ ویئر اور نظام ہیں جو معذور افراد کی مدد کرتے ہیں تاکہ وہ اسکول، کام اور معاشرے میں برابر حصہ لے سکیں۔ ان کی چار اقسام ہیں: نقل و حرکت میں مدد (اسمارٹ ویبل چیئر)، حسی مدد (سماعتی آلہ، اسکرین ریڈر، بریل ڈسپلے)، گفتگو میں مدد (اسپیچ جنریٹنگ ڈیوائس) اور ذہنی و تعلیمی مدد (آڈیو بکس، اسپیل چیکر، یاد دہانی ایپس)۔

EXAM TIP

Expect: "What are assistive technologies? Explain their categories with examples" (long), "Name two sensory aids" (short), or "A screen reader is an example of ____" (MCQ).

For full marks write the definition (devices, software and systems; help people with disabilities; equal participation), then all **four** categories each with purpose + two examples. Finish with one line on importance (equal access, independence, inclusion).

CHECK YOURSELF

1. Which category does a hearing aid belong to?
2. Give two examples of cognitive/learning aids.
3. Define assistive technologies in one sentence.

Answers: 1. Sensory aids. 2. Any two: audiobooks, spell-checkers, digital reminder apps. 3. Devices, software and systems that help people with disabilities or functional limitations participate equally in school, work and society.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Assistive technology helps:

- a. People with disabilities use systems
- b. Only hackers
- c. Only DBMS keys
- d. Only Agile coaches

MCQ 2. A screen reader is for:

- a. Blind / low-vision users
- b. Faster bubble sort
- c. Wi-Fi
- d. Primary keys

MCQ 3. Captions help:

- a. Deaf / hard-of-hearing users (and many others)
- b. Routers
- c. NAND gates
- d. Pie charts only

MCQ 4. Importance: they support:

- a. Equal access to education and work
- b. Faster CPUs only
- c. Cheaper RAM only
- d. OSI layer 1 only

MCQ 5. A career path is:

- a. Accessibility specialist / AT developer
- b. Only analog radio
- c. Only waterfalls
- d. Only K-maps

MCQ 6. Good HCI + AT means:

- a. Design for a wide range of users from the start
- b. A separate broken version forever
- c. No keyboard needed ever
- d. No contrast ever

Extra short questions

1. Define assistive technology with two examples.
2. Why it matters in a Pakistani college.
3. Link to HCI accessibility principles.
4. One career connection from the book.

Extra long question

1. Propose AT + ordinary design changes so a student with low vision and a student with limited hands can use the college portal. 10 marks. ★ if tagged Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Devices/software that help functional limits: screen readers, switches, captions, magnification.
2. A blind student can still take CS if materials are readable by NVDA and papers have electronic copies.
3. Perceivable, operable, understandable, robust — AT relies on those.
4. AT designer, accessibility tester, special-education technologist.

Long

1. Low vision: contrast, zoom, screen reader labels. Limited hands: large targets, keyboard/switch, extra time. Test with real users. Do not ship a mouse-only form.